AFZAL KHAN

Senior AI Engineer | Post-Training, Evaluation, Agentic Systems, MLOps & Formal Methods
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PROFESSIONAL SUMMARY

Senior AI engineer with 4+ years across machine learning, LLM post-training, evaluation infrastructure, agentic systems, computer vision, data engineering and cloud deployment. Leads an AI engineering team at ClaimbAI and owns post-training/evaluation workflows spanning SFT, GRPO/PPO, reward models, formal verifiers and multi-step tool-use scoring. Background in mathematical logic and New Foundations set theory complements an engineering focus on precise specifications, edge cases, failure modes and mechanically checkable correctness.

CORE COMPETENCIES

- LLM post-training & RL: SFT, GRPO, PPO, preference pairs, reward-model training, reward shaping, RLVR-style verifier signals, process/milestone rewards, credit assignment, reward-hacking analysis, TRL, PyTorch, Hugging Face, vLLM.
- Evaluation & formal verification: gold-solution authoring, weighted rubrics and partial credit, held-out splits, multi-turn rollout scoring, verifier design, regression testing, task specifications, property-based testing, failure-mode analysis.
- Agentic AI & RAG: tool/function calling, MCP clients/servers, multi-agent orchestration, LlamaIndex, LangChain/LangGraph, pgvector/PostgreSQL, n8n, voice agents, local SLMs/Ollama.
- MLOps, cloud & data: Kubernetes, Docker, CI/CD, MLflow, FastAPI/vLLM serving, GPU scheduling, Redis, PostgreSQL, Snowflake, Databricks, AWS, Azure, GCP, Vercel, AWS Glue.
- Computer vision & edge AI: OpenCV, YOLO v9-v11, TensorRT, ONNX, quantisation, pruning, NVIDIA Jetson Orin, ARM64/aarch64, tracking and real-time C++/Qt systems.

PROFESSIONAL EXPERIENCE

ClaimbAI - Senior AI Engineer — Jul 2025 - Present

Remote - California, USA | Team lead across post-training, evaluation, agentic systems and AI infrastructure.

- Own the end-to-end post-training loop for an in-house SLM: supervised fine-tuning, preference-based optimisation with GRPO and PPO via Hugging Face TRL, reward-model training, rollout collection and dataset versioning; serve models through vLLM.
- Designed formal verifiers and reward shaping for multi-step tool-use trajectories, decomposing long tasks into milestone checks so partial progress is scored while tool selection, argument correctness and recovery from failed calls remain auditable.
- Built the internal AI evaluation platform from scratch: task/environment definitions, gold solutions, rubric-based partial-credit scoring, held-out splits, trajectory inspection and checkpoint comparisons; tightened task sets against shortcut strategies and metric gaming.
- Architected production RAG infrastructure using PostgreSQL/pgvector and LlamaIndex, including indexing, chunking and embedding strategy; built and maintained 12+ secure connectors including Snowflake, Databricks, AWS, Azure, GitHub and Slack with OAuth/RBAC.
- Designed multimodal voice-agent workflows and local/private SLM retrieval systems; contributed full-stack and platform engineering across APIs, web/mobile integrations and secure enterprise data ingestion.
- Own deployment and platform work across AWS, Azure, GCP and Vercel: Kubernetes, Docker, GPU scheduling, CI/CD, monitoring, model serving and quality/drift tracking across model versions and prompt revisions.

COMCEPT - Software Engineer — Jan 2025 - Oct 2025

Islamabad, Pakistan

- Engineered real-time computer-vision and multi-object-tracking stacks (YOLOv9-v11, GOTURN, CSRT, DASiamRPN/SAMURAI and custom models) with optimised inference on NVIDIA Jetson AGX Orin/Orin NX.
- Developed C++/Qt/QML pipelines with multithreading, CUDA/OpenCV and TensorRT; model optimisation through pruning and quantisation improved inference throughput by about 30% and reduced false positives by about 25%.
- Built telemetry/monitoring interfaces and integrated firmware, ARM64/Linux compute nodes and networked devices under hard latency, memory, power and reliability constraints.

AIGENIUS - Chief Operating Officer — Jul 2024 - Dec 2024

Islamabad, Pakistan

- Directed operations, recruitment, client delivery and engineering execution while remaining hands-on technically.
- Led a 200-million-row transformation across roughly 200 tables into a normalised 3NF warehouse and star-schema semantic layers for analytics, forecasting and dashboards; implemented ETL using AWS Glue and SQL.
- Built a Node.js/React agent chatbot, web-scraping tooling and a football player-tracking proof of concept; managed client relationships and technical delivery.

Convergent Business Technologies - Consultant, Analytics & Insights — Aug 2023 - Jul 2024

Islamabad, Pakistan

- Automated the Pepsi KSA/UAE analytics programme, reducing refresh time from one week to three days and required staffing from six people to two through query tuning and scripting.
- Built ETL, 3NF/dimensional models, semantic layers and Tableau/Power BI analytics; developed an ML/audio pipeline for pilot-tower communications with audio cleaning, ASR, OCR/timestamp extraction and post-processing.

Upwork - Freelance Data / ML Engineer — Aug 2022 - Aug 2023

Remote

- Delivered end-to-end analytics and predictive systems for telecommunications, healthcare and finance: churn, forecasting, inventory optimisation, feature engineering, model training, ETL and deployment.

- Built SQL/AWS Glue data workflows and Tableau/Power BI reporting; trained client teams and supported analytics adoption.

National Centre of Robotics and Automation (NCRA) - Research Intern Jun 2021 - Aug 2021

Islamabad, Pakistan

- Developed a MATLAB GUI for microrobot movement analysis and real-time C++/OpenCV image-processing software; supported collection, storage and analysis of robotics experimental data.

RESEARCH IN MATHEMATICAL LOGIC AND SET THEORY

Independent research preparation - New Foundations (NF/iNF) Jul 2026 - Present

Guided by Dr Thomas Forster; research direction focused on proof-theoretic consistency of intuitionistic New Foundations.

- Worked through a fourth-year mathematical logic course with proofs and exercises covering ordinals, cardinals, wellfoundedness, fixed-point theorems, first-order logic, compactness, axiomatic set theory and model theory.
- Completed and submitted Forster's stratification problem set, then expanded selected solutions into expository discussions; studied NF permutation models, typical ambiguity, the Holmes-Wilshaw Con(NF) blueprint/deformalisation, and related primary literature.
- Currently working through Gentzen cut elimination and the epsilon_0 consistency machinery, with the proposed research problem centred on an iNF consistency proof and the obstruction to transfer to classical NF.
- Technical correspondence with Prof. Murdoch J. Gabbay on stratification/rewriting and Thomas Porter on symmetry-principled type theory; study of Lean 4 formalisation and proofs associated with Con(NF).

EDUCATION

MITx / edX - MicroMasters Program in Statistics and Data Science In progress

Capstone remaining (Fall 2026).

- Verified courses completed: Probability - The Science of Uncertainty and Data (6.431x); Fundamentals of Statistics (18.6501x); Machine Learning with Python (6.86x); Learning Time Series with Intervention (IDS.S24x).

National University of Sciences and Technology (NUST) - BE Mechatronics Engineering Sep 2019 - Jun 2023

College of Electrical & Mechanical Engineering | CGPA: 3.27/4.00

- Coursework includes programming, data structures/OOP, circuits, signals and systems, digital logic, calculus, linear algebra/ODEs, vector calculus, complex variables/transforms, probability/statistics, numerical methods, modeling/simulation, control, robotics, embedded systems, mechanics, thermodynamics and automation.
- Final Year Project: Autonomous Multipurpose Agri-bot - designed/manufactured a modular agricultural robot with seeding/drilling modules, autonomous navigation/object avoidance and a mobile control application; A grades in FYP I and FYP II.

AWARDS AND RECOGNITION

- NCRA Best Design Award - Indigenous Category, for a mini autonomous fruit-plucking and navigation robot competition.
- Final-year engineering project recognition through NUST/NCRA robotics and agricultural automation work.

PRODUCT, RESEARCH AND ENGINEERING PROJECTS

AI Evaluation Platform

Self-built evaluation harness for agent/model runs with pluggable task definitions, gold solutions, weighted rubrics and partial credit, held-out task splits, trajectory inspection, checkpoint comparison and regression testing.

Quanta - Legal Valuation Engine

Solo-built deterministic engine for UK personal-injury claims quantum that produces a defensible valuation, exposes calculation logic and supports associated document drafting; deployed using Vercel/Modal.

Grip Intelligence AI - Golf Swing Coach

Mobile-first golf coaching product using high-frame-rate swing capture and pose/biomechanics measurement, with reliability gates, metric grading and visual feedback; Android/iOS deployment work and production backend integration.

ArchitectAI / Forma

Constraint-driven residential floor-plan generator taking plot geometry, dimensional/floor constraints and preferences to generate compliant layouts and exportable architectural artefacts such as DXF/DWG/PDF.

Care Companion

AI-assisted health intake/triage MVP for elderly users with voice/text intake, follow-up questions, interim guidance, structured recommendation reports, doctor routing and patient/doctor/admin workflows.

Ciara Ride-Hailing Platform

Backend and real-time socket architecture for car, bike, shared rides and food workflows; geospatial driver matching with MongoDB, driver rooms, booking pipeline and load-testing of socket flows.

Voice AI Receptionist

End-to-end phone agent pipeline combining speech-to-text, LLM reasoning/tool use and text-to-speech, with telephony/SIP integration experiments and local testing.

Multi-Agent Business Automation

Central controller routing work across 10-15 specialised agents per department (Sales, HR, Marketing and Engineering), with retries, state handoff and workflow automation.

Local Research AI Agent

Local agent built with n8n and Ollama, web-search integration, PostgreSQL vector retrieval and persistent memory across sessions.

Model Context Protocol Client
Local MCP client for scoped, permissioned filesystem access; hands-on implementation of tool schemas, permissioning, error handling and agent failure modes.

Advanced Sports Performance Analytics
Computer-vision platform for player tracking and movement/performance metrics, predictive analysis and interactive coaching/management dashboards.

Adaptive Language Learning Web App
Firebase/Gemini application generating personalised lessons, conversational practice, grammar feedback, progress tracking and course adaptation.

Face Recognition Software
MATLAB/OpenCV facial-recognition pipeline using transfer learning on AlexNet; 92% classification accuracy on a custom dataset with real-time capture/processing.

Autonomous Multipurpose Agri-bot
NUST final-year project: CAD/ANSYS mechanical design, modular agricultural tooling, sensor integration, autonomous movement/object avoidance and mobile control.

6-DOF Robotic Manipulator
Designed and fabricated a 6-degree-of-freedom robotic arm with CAD, high-torque servos, gears, acrylic structure and Arduino-based control.

Advanced Data Warehouse Transformation
Architected a 3NF data warehouse from approximately 200M rows across roughly 200 tables, with star-schema analytics layers and automated ETL via AWS Glue/SQL.

TECHNICAL SKILLS

Programming	Python (advanced), C++, SQL, TypeScript/Node.js, React, Bash, MATLAB, Rust, Qt/QML, Git, Linux.
AI / LLMs	PyTorch, Hugging Face, TRL, vLLM, transformers, RAG, LlamaIndex, LangChain/LangGraph, Gemini, local SLMs/Ollama, voice agents, tool/function calling, MCP.
Post-Training & Evaluation	SFT, GRPO, PPO, preference data, reward-model training, verifier design, milestone/process rewards, rubric design, gold solutions, held-out evaluation, rollout scoring, checkpoint regression.
Computer Vision / Edge	OpenCV, YOLO v9-v11, TensorRT, ONNX, quantisation, pruning, tracking, Jetson Orin, CUDA-enabled pipelines, ARM64/aarch64.
Data & Analytics	PostgreSQL/pgvector, Redis, Snowflake, Databricks, SQL, ETL, 3NF and dimensional modelling, AWS Glue, Tableau, Power BI, pandas, NumPy, scikit-learn.
Cloud / MLOps	Docker, Kubernetes, CI/CD/GitHub Actions, FastAPI, MLflow, AWS, Azure, GCP, Vercel, Modal, GPU scheduling, monitoring and model serving.
Formal Methods	Lean 4 study, specification of machine-checkable correctness conditions, verifier design, property-based testing, regression testing, LaTeX.
Engineering / Robotics	CAD/ANSYS, embedded systems, robotics, sensors, control, automation, system integration, telemetry and real-time software.

CERTIFICATIONS AND ADDITIONAL TRAINING

- CS50 - Harvard University; Google Data Analytics - Coursera; Advanced SQL for Data Analysis - Udacity; McKinsey Forward Program; Tableau Data Analysis; UI/UX Design - CalArts.
- MITx graduate-level statistics and data-science coursework listed under Education; additional self-directed study in mathematical logic, set theory, formalisation and proof theory.

LANGUAGES

- English - fluent (medium of instruction throughout education); Urdu - native; Pashto - fluent/conversational; Arabic - reading.

REFEREES

Dr Anas Bin Aqeel
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Taha Rasheed
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